

the diode is abruptly switched from forward-biased condition to reverse-biased condition, the time required by the reverse current or voltage to reach a specified value is called 'reverse recovery time.' The reverse recovery time is a very significant parameter in fast-recovery rectifiers used in SMPS and in diodes used for high-frequency switching applications. In fact, it is the most important parameter to watch while choosing a rectifier diode for an SMPS based on the flyback principle.

The reverse recovery time varies from a few nanoseconds for ultra-fast diodes to about 500 nanoseconds for a typical fast-recovery rectifier. For example, reverse recovery time for ultra-fast diode 1N914 is of the order of 4 ns, while it is around 300 ns for fast-recovery rectifier BY299. The reverse recovery time parameter of the rectifier when used in a flyback-type DC-to-DC converter puts an upper limit on the maximum usable switching frequency.

Forward recovery time (t_f). It is the time required for the forward current or voltage to reach a specified value after the diode has been switched from reverse-biased state to forward-biased state. This parameter too is significant in switching applications.

Thermal characteristics

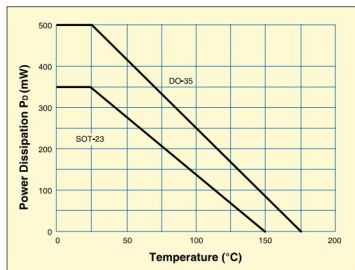


Fig. 9: Power dissipation—temperature derating curve of 1N914

Parameters defining thermal characteristics of the device are intimately related to one another. Power dissipation capability is a function of ambient temperature. It is also related to the junction-to-case thermal resistance and the maximum allowable junction temperature. These parameters should therefore not be seen in isolation.

Power dissipation and thermal resistance are two important thermal parameters. The designers should not look at these parameters in isolation. They need to be considered together to ensure that the diode's maximum allowable junction temperature specification doesn't exceed under the operating conditions of the diode's forward voltage, forward current and ambient temperature. We shall first define the two parameters and then illustrate their inter-relationship with actual data quoted from a datasheet.

Power dissipation (P_o). It is the maximum power that can be safely dissipated in a diode for a given ambient temperature in the absence of any heat sink. The value of maximum power dissipation is usually specified at an ambient temperature of 25°C. At higher operating temperature, its value should be derated as per the power-temperature derating curve of the diode. As an illustration, the power derating curve of 1N914 is shown in Fig. 9. The maximum

power rating decreases linearly with the increase in temperature at a rate of 3.33 mW/°C above 25°C.

Thermal resistance from junction-to-ambient (θ_{JA}). This parameter is an indicator of the diode's

junction-to-ambient temperature differential for a known value of power dissipated in the device. It is expressed in °C/watt. For given values of power dissipation, ambient temperature and junction-to-ambient thermal resistance of the device, one can estimate the junction temperature. If the device has to safely dissipate the specified power at an ambient temperature greater than the power dissipation rating is specified for, it should be mounted on a heat-sink to reduce its junction-to-ambient thermal resistance.

Referring to the datasheet of 1N914, the specified power dissipation rating is 500 mW at an ambient temperature of 25°C and its maximum junction temperature specification is 175°C. This is in agreement with its thermal resistance specification, which is specified to be 300°C/watt. As per this specification, there will be a temperature differential of 150°C between ambient and junction for a power dissipation of 500 mW. This means that the junction temperature would be 175°C.

If this device has to safely dissipate 500mW power at, say, 75°C, the temperature differential cannot exceed 100°C. This would necessitate a thermal resistance from junction to ambient of 200°C/watt. It can be achieved by using a suitable heat-sink, thus reducing case-to-ambient thermal resistance component of the overall thermal resistance.

...To be concluded next month

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